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Basin infill architecture and evolution from magnetostratigraphic cross-basin correlations in the southeastern Pyrenean foreland basin

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ABSTRACT

Magnetostratigraphy, numerical dating, and facies mapping have been combined to provide the first cross-basin correlation of the Eocene southeastern Pyrenean foreland basin in northeastern Spain. This has enabled (1) depositional systems on the northern and southern margins of the basin to be dated and correlated across the basin and (2) changes in the rates of sediment accumulation in time and space to be evaluated. By sampling correlative intervals at several locations in a transect across the basin, magnetozones were detected that otherwise may have been misinterpreted due to recent magnetic overprinting, or missed by wide sampling intervals in previous magnetostratigraphic surveys. Results of this study indicate that marginal marine strata of the southeastern part of the foreland near the Vic area of Spain are 5 m.y. older than previously thought, the base of the interval (46 Ma) being middle Lutetian in age. By proposing an age of at least 35.68 Ma for the overlying Cardona evaporite unit, the age span of the marine interval in the studied part of the foreland is ~10 m.y. Contrary to earlier interpretations, no major southward migration of the southeastern Pyrenean foreland basin depocenter is recorded by the basin stratigraphy during the initial stages of middle Eocene marine sedimentation in the southern margin of the basin. However, marked basin asymmetry is observed later in its evolution as the top of the marine deposits records northward increase in the thickness of chron C17n.1n (37.47–36.61 Ma). This asymmetry is also proved by the presence of a larger number of magnetozones within marine strata at the top of the marine interval in the northern sector. These results imply that previous age calibrations and the estimated rates of tectonic processes and sedimentation within the foreland need to be reassessed. The implications for basin infill architecture evolution through time and space, subsidence analysis, and depocenter evolution through time are addressed.

INTRODUCTION

Quantitative studies of processes controlling the infill architecture of sedimentary basins require reliable, basinwide age control. Precise dating is often made difficult by structural complexities, poor outcrops, and facies changes that occur over relatively short distances, the latter being particularly evident in foreland basins. Marginal marine deposits in foreland basins commonly include shallow-water facies associations that are either barren of fossils or contain only benthic fauna. Stratigraphic analysis based on dating by magnetic polarity reversals may help overcome this problem. However, because magnetostratigraphic dating of sedimentary rocks requires complementary age constraints, sections are commonly selected that provide the most extensive biostratigraphic information (e.g., Serra-Kiel et al., 1994), i.e., those corresponding to the deepest sectors of the basin. In contrast, good age constraints are in general lacking in the marginal deposits of sedimentary basins, rendering correlation across a basin practically impossible. However, this problem is beginning to be addressed by the magnetostratigraphic correlation of several distinct sections across a basin (e.g., Bentham et al., 1992; Bentham and Burbank, 1996; Steiner et al., 1993).

Previous studies in the Ebro foreland basin have focused on dating part of the allochthonous units in the Jaca-Tremp area (Bentham et al., 1992; Hogan and Burbank, 1996) and on complementing the results from magnetostratigraphy with the available biostratigraphic data of planktonic and benthonic forams studied at the same sections (Pascual, 1992; Pascual et al., 1992; Serra-Kiel et al., 1994). Burbank et al. (1992a, 1992b) used magnetostratigraphy to analyze the changes in subsidence patterns between the external thrust sheets and the hinterland in the easternmost sector of the Ebro foreland basin (herein referred to as southeastern Pyrenean foreland basin). However, evidence of Eocene thrusting in the hinterland of the Pyrenees (Berastegui et al., 1996) implies that cross-basin correlation is required to unravel the effects that both sets of thrusting (and their interaction) have had on basin evolution and infill architecture.

The easternmost southeastern Pyrenean foreland basin (Fig. 1) was chosen for this study due to the proximity of the two orogenic fronts. This is the only area of the south Pyrenean foreland where both north- and south-sourced Eocene marine platform depositional systems are exposed. The aim of this paper is to examine by basinwide magnetostratigraphic and lithologic correlation the Eocene marginal marine sequences of the foreland that are between two fold-thrust belts (Pyrenees and Catalan Coastal Ranges). These belts were converging throughout the evolution of the basin (Vergés

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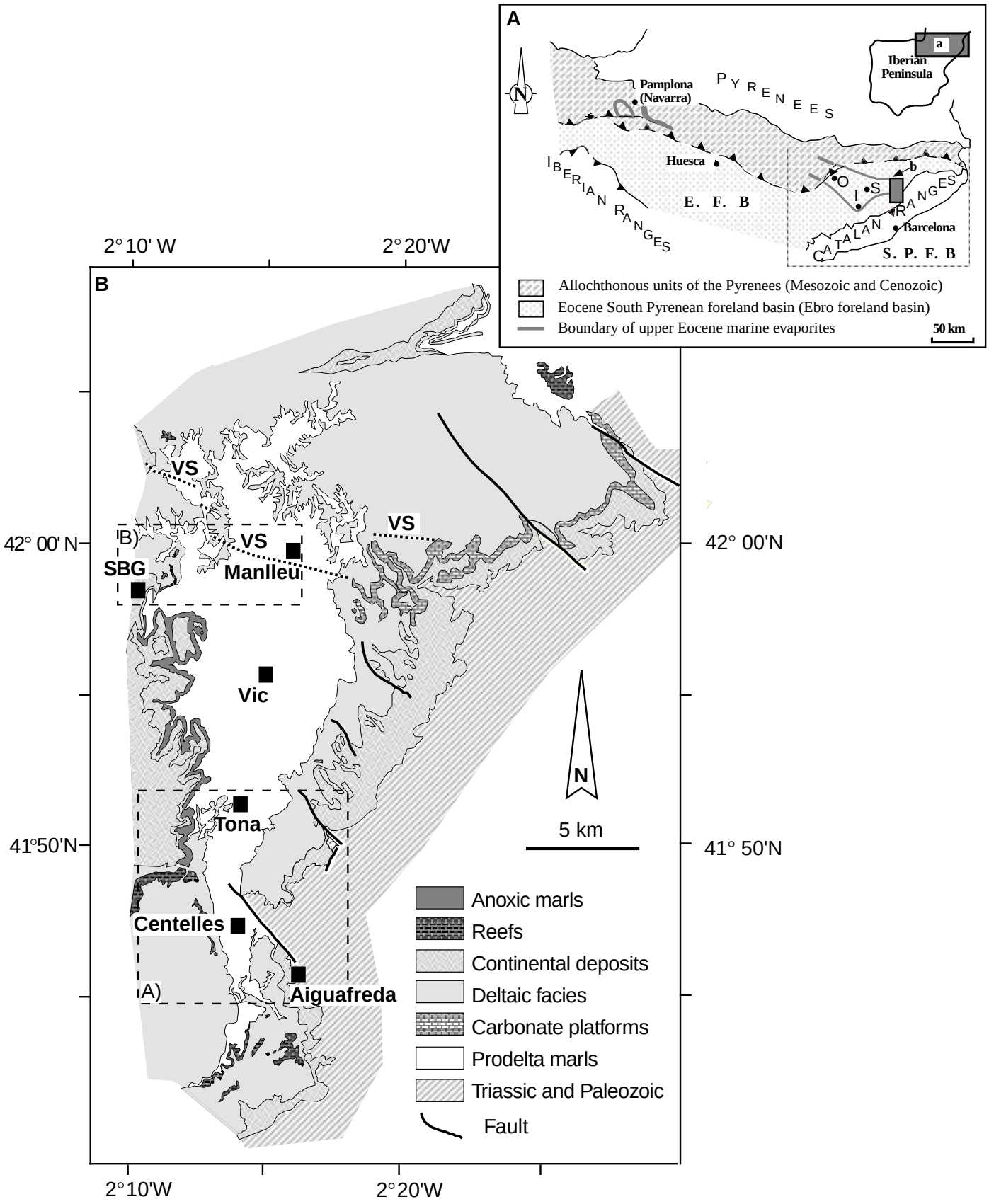


Figure 1. General location map of northern Spain (A) and general facies map of the study area (B), the Vic area of the Catalan sector of the southeastern Pyrenean foreland basin (boxed area in A). Boxes A and B (in B) show Figure 2. EFB—Ebro foreland basin, SPFB—southeastern Pyrenean foreland basin, O—Oliana, S—Suria, I—Igualada, SBG—Sant Bartomeu del Grau, VS—location of Vic section from Burbank et al. (1992a).

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et al., 1995; Berastegui et al., 1996). In this study numerical dating ($^{49}Ar/^{39}Ar$, $^{87}Sr/^{86}Sr$) constrains the results of the magnetic polarity reversal determination, and adds an independent age calibration of the geomagnetic polarity time scale (GPTS). This represents the first cross-basin study of its type in the basin and throws light on how both margins evolved with respect to one another. Furthermore, we provide a new chronology for the southeastern Pyrenean foreland basin, which will serve as the framework for future studies in the foreland (e.g., tectonics, basin modeling, as well as form the basis for a reassessment of subsidence analyses).

GEOLOGIC SETTING AND CONSTRAINTS FROM PREVIOUS BIOSTRATIGRAPHIC AND PALEOMAGNETIC STUDIES

The Ebro foreland basin is part of the foreland basin system of the Pyrenean orogen, formed as the product of a complex deformation sequence as the Iberian microplate both moved past, and collided with, the European plate. This occurred principally between Late Cretaceous and Oligocene time (e.g., Puigdefàbregas and Souquet, 1986). Outcrops of Paleogene rocks from this foreland (Fig. 1A) are present in thrust sheets in the north, and in autochthonous strata bounded by orogenic belts of the Iberian Range in the south and southwest and the Catalan Coastal Ranges in the southeast. The southeastern Pyrenean foreland basin (Bentham et al., 1992; Vergés et al., 1998) corresponds to the eastern side of the Ebro foreland basin developed mainly on pre-Mesozoic basement that forms an irregular triangular shape bounded by the Pyrenees to the north and the Catalan Coastal Ranges to the southeast. The sections sampled for magnetic polarity reversal studies are located at the eastern side of the southeastern Pyrenean foreland basin (Figs. 1B and 2). The sedimentary rocks that crop out in this area are Paleocene to Oligocene in age and consist of continental deposits that are below and above marine strata. These marine sediments are traditionally considered to be middle to upper Eocene in age (e.g., Hottinger, 1960; Ferrer, 1967, 1971; Reguant, 1967; Via, 1969; Serra-Kiel, 1984). The sedimentary formations distinguished by several authors among the marine Eocene deposits of the Vic sector of the southeastern Pyrenean foreland basin are shown schematically (Fig. 3) after the synthesis proposed in Taberner (1983).

The base of the marine sequence has been interpreted as marking a rapid late Lutetian to Bartonian transgression that occurred as a result of the emplacement of the floor thrust of the Pyrenean antiformal stack (Puigdefàbregas et al., 1986). East of Vic, the basal marine strata have been dated as middle Lutetian age (Serra-Kiel, 1984). However, the first marine deposits in the southern sector of the basin were interpreted as lower-middle Biarritzian age (included in the Bartonian in present stratigraphic scales) by Reguant (1967). This implies a strong diachroneity of the base of the marine deposits, on the basis of which Puigdefàbregas et al. (1986) proposed southward migration of the south Pyrenean depocenter due to the emplacement of the south Pyrenean thrust sheets. Serra-Kiel and Travé (1995) and Vergés and Burbank (1996) interpreted the lowermost marine sediments cropping out at the southern margin of the southeastern Pyrenean foreland basin to be diachronous and to span from middle Lutetian time to the base of chron C18r at 41.2 Ma (Bartonian).

The top of the marine section was considered to be lower Priabonian age based on (1) nummulite species present beneath the youngest reef deposits (Serra-Kiel, 1984) and (2) the micromammal assemblage and charophyte biozonation in the continental sediments above the marine interval, that suggest an upper Eocene–Oligocene age (Agustí et al., 1987; Anadon et al., 1987; Feist et al., 1994). Burbank et al. (1992a) linked the top of their magnetostratigraphic section from the Vic sector of the basin and the evaporites equivalent to the Cardona Formation to the significant sea-level fall that occurred at the Bartonian-Priabonian boundary (Haq et al., 1987), and gave an age of around 40 Ma for the evaporites of the Cardona Formation (located in Fig. 3). However, Vergés and Burbank (1996) and Serra-Kiel et al. (1998) linked these evaporites to the base of the Priabonian (37.2 Ma) after reevaluation of the original magnetostratigraphic data (Burbank et al., 1992a) against the revised GPTS produced by Cande and Kent (1995).

NUMERICAL DATING

The age for the base of the marine sequence has been constrained by $^{40}Ar/^{39}Ar$ dating of the lowermost marine deposits (glauconite-rich marls). The glauconite-rich marls are found just above the lower continental deposits and transgressive beach conglomerates in the sector of Aiguafreda (Fig. 1B and sample RV-53 in Fig. 2A). The glauconite grains are well preserved and show no evidence of transport after their formation or alteration by recent weathering. The potassium content (6.7% K_2O) and the shape (irregular rounded grains with cracks) of these glauconites suggest that they are authigenic grains, formed in a marine environment very close to the sediment-water interface, and thus they are classified as evolved-type glauconites (J. Solé, 1995, personal commun.) in the classification scheme of Odin and Fullagar (1988). These features imply that the glauconites are of suitable quality for the purposes of numerical dating (Odin and Dodson, 1982).

An approximation of the age for the evaporites (Cardona Formation, Fig. 3) is obtained from $^{40}Ar/^{39}Ar$ dating of limpid sylvite crystals present in the basinal potash beds in the Suria mine (Fig. 1A). A further constraint for the age of this evaporite formation is obtained from $^{87}Sr/^{86}Sr$ of sulfates intercalated in the evaporite sequence below the potash beds.

The samples (one glauconite and one sylvite) were processed and analyzed by $^{40}Ar/^{39}Ar$ dating at Princeton University Laboratories (supervised by T. C. Onstott). The samples were enclosed in a finger-glove under vacuum and irradiated for two days. Gas was then obtained by heating under vacuum by laser (step heating for the glauconite sample). For the glauconite sample, the obtained integrated age, instead of the plateau age, has been used as a reference for our magnetostratigraphic data. Although the sample gave a reasonable plateau containing more than 50% Ar gas released during heating (see Table 1), Smith et al. (1993) proved that glaucony minerals have recoiling of both ^{37}Ar and ^{39}Ar atoms, causing plateau ages anomalously older than integrated ages. Nevertheless, these authors suggest that the integrated ages agree favorably with conventional K-Ar results. The integrated age obtained for the glauconite sample is 46.8 ± 0.8 (2σ) Ma (Table 1). This implies that the base of the marine strata at the southern margin of the basin in the Vic area is much older than previously proposed (41.25 Ma, Serra-Kiel and Travé, 1995; Vergés and Burbank, 1996).

During heating, the sylvite sample tended to sublimate rather than degas. As a consequence there is no clear pattern in the Ar release during heating to provide a plateau; the integrated age has thus been chosen for an approximation of the numerical age of the evaporite unit. The integrated age of the sylvite sample is 37.06 ± 0.8 (2σ) Ma (Table 2). This result may be liable to a larger error than the integrated age obtained for the glauconite sample, due to the analytical problems discussed here (sublimation of the sample rather than degassing).

Recently obtained $^{87}Sr/^{86}Sr$ analyses (Table 3) from anhydrite samples provide further age constraints on the evaporite formation. The $^{87}Sr/^{86}Sr$ analyses were processed at the University of London Radiogenic Isotope Facility at the Department of Geology, Royal Holloway–University of London, under the supervision of Matthew Thirlwall. The numerical age obtained, using the $^{87}Sr/^{86}Sr$ look-up table of Howarth and MacArthur (1997), suggests a younger age (35.1 ± 0.4 to 33.8 ± 0.2 Ma) for the evaporite formation than that obtained from $^{40}Ar/^{39}Ar$ analysis of sylvite or that previously assumed by Burbank et al. (1992a) after correlation to the sea-level curve provided by Haq et al. (1987).

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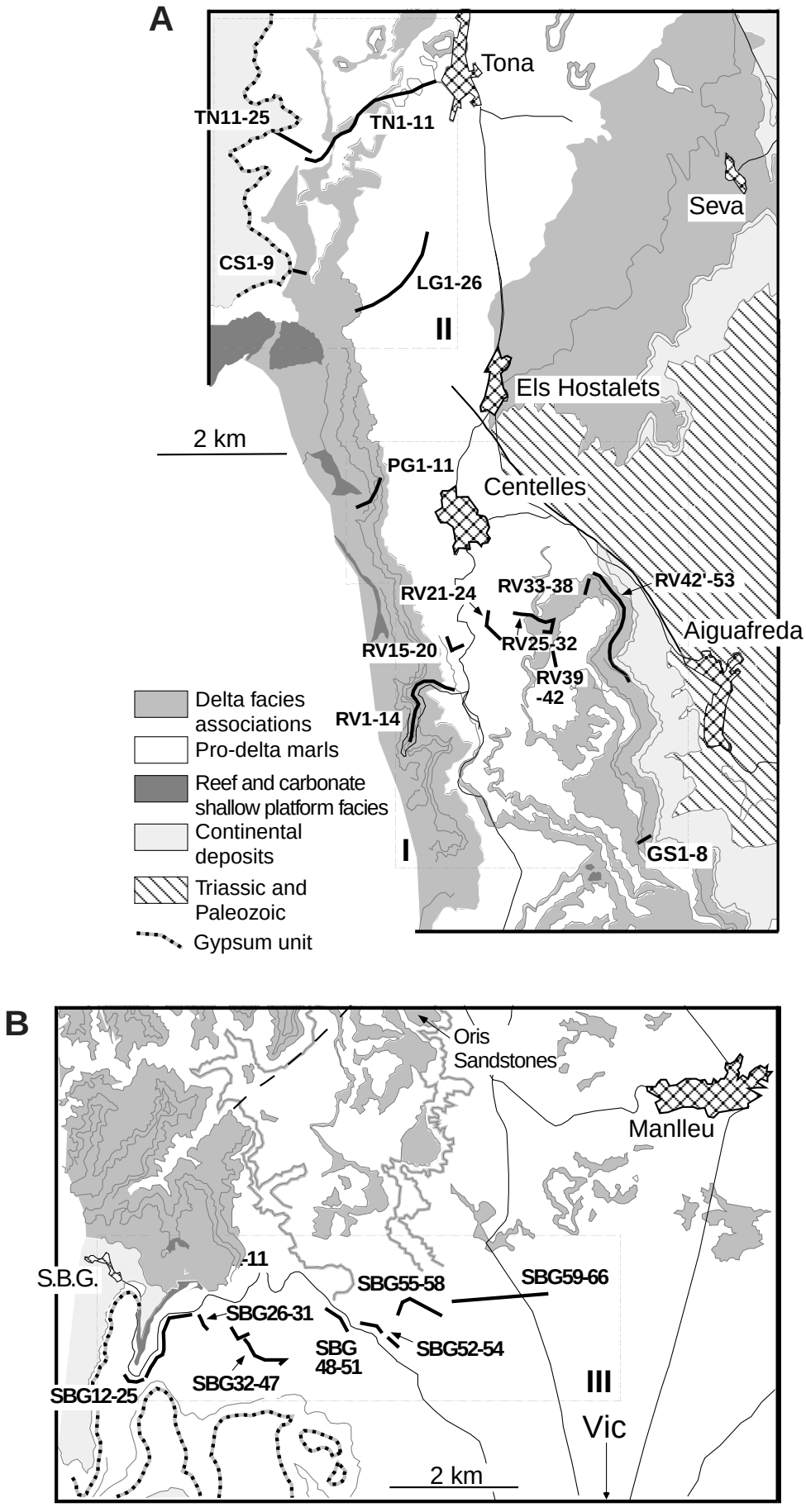


Figure 2. Maps displaying the location of sampling sites in the south (A) and the north (B) of the Vic sector. SBG—Sant Bartomeu del Grau village. I, II, and III show the three sectors in which the paleomagnetic sections have been grouped for statistical purposes. Heavy black lines indicate sections sampled for paleomagnetism.

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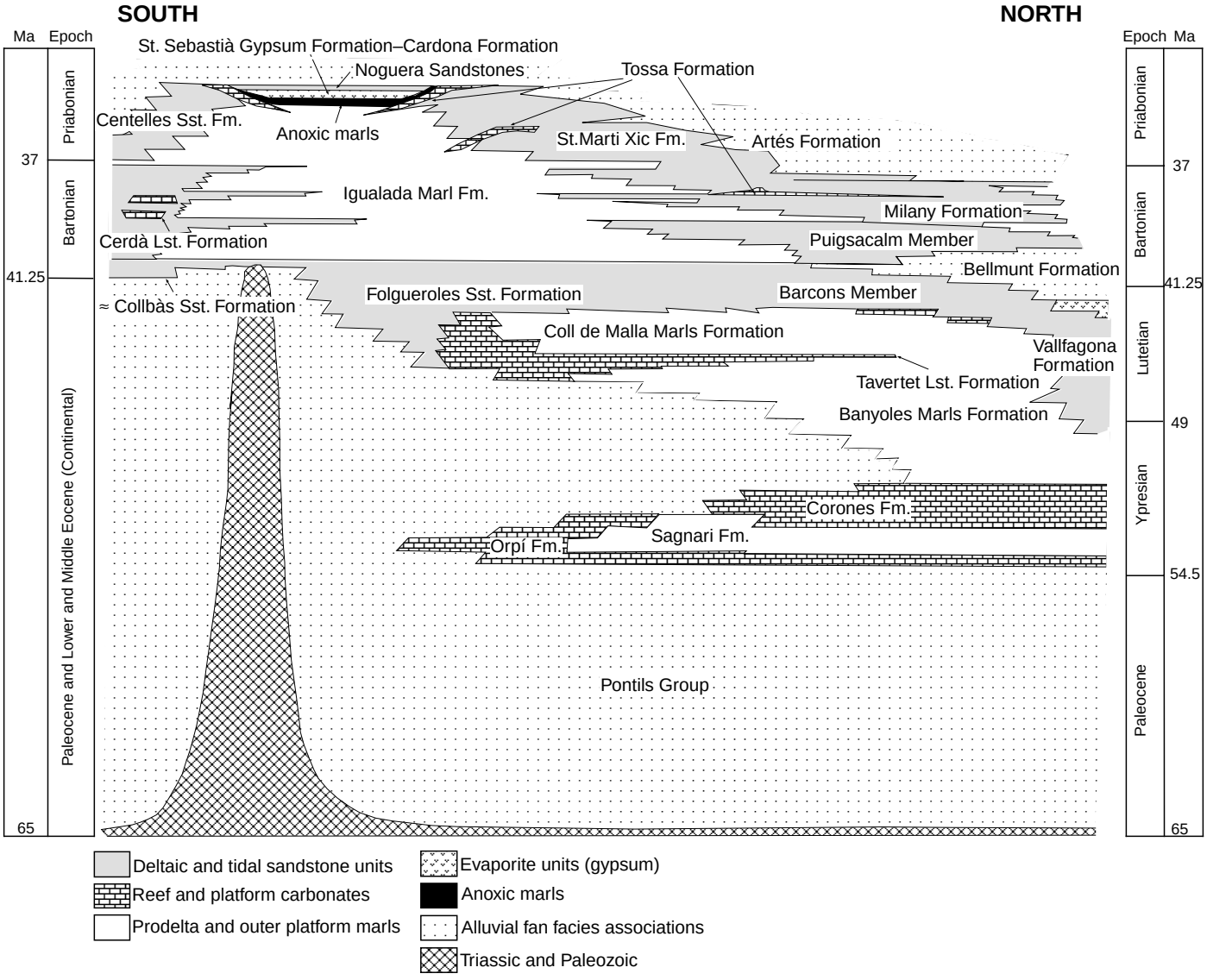


Figure 3. Chronostratigraphic diagram for the Vic sector of the southeastern Pyrenean foreland basin. Scheme from Taberner (1983) is modified for time scales according to Cande and Kent (1995) and Berggren et al. (1995). This scheme synthesizes information published before this study and the sedimentological interpretations of deposits filling the basin (Taberner and Santisteban, 1987; Santisteban and Taberner, 1988a, 1988b; Pierre et al., 1994). Puigdefàbregas et al. (1986) distinguished eight sequences of basin infill, the three topmost sequences of which include the following formations from the northern margin of the basin: (1) the Bellmunt sequence, including the Tavertet, Banyoles, Barcons, and Bellmunt Formations; (2) the Milany sequence, including the Puigsacam Member, and Milany, Folgueroles, Collbàs, Tossa, and Igualada Formations; (3) the Cardona sequence, including the St. Sebastià Gypsum Formation and Cardona Formation.

The numerical age for the evaporite formation provided from the two dating methods yields a difference of at least 2 m.y. This difference can be explained by the imprecision of the methods and the analytical problems involved with gas release of the sylvite sample during $^{40}Ar/^{39}Ar$ analysis. The agreement of results from $^{87}Sr/^{86}Sr$ analysis of sulfates from the lowermost anhydrite unit and sulfates dispersed in the halite unit above (Table 3) suggests that this method might better approach the age of the formation for these particular mineralogies. Conclusions on the age of the evaporite formation at the top of the marine sequence are discussed later when comparing with the results of correlation of the topmost part of the marine sequence to the magnetic polarity scale.

MAGNETOSTRATIGRAPHIC SAMPLING AND METHODOLOGY

Seven sections in the Vic area of the southeastern Pyrenean foreland basin were sampled and measured. These include the La Rovira, El Grau del Sunyer, Puigsagordi, La Guardia, Collsuspina, Tona, and Sant Bar-

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TABLE 1. $^{40}Ar/^{39}Ar$ RESULTS OF LASER STEP HEATING OF THE GLAUCONITE SAMPLE

Step	Cumulative ^{39}Ar	Age (Ma)	±
1	0.0002	–448.86	652.99
2	0.0040	14.84	16.95
3	0.0257	4.78	5.79
4	0.0260	93.91	84.71
5	0.0590	18.40	2.23
6	0.0719	45.23	4.00
7	0.1259	40.65	1.46
8	0.1877	50.04	1.67
9	0.3555	48.40	0.77
10	0.4107	52.65	1.05
11	0.4335	45.66	2.80
12	0.4906	47.96	0.89
13	0.5951	47.75	0.98
14	0.8135	49.91	0.50
15	0.9518	50.35	0.81
16	0.9928	50.06	1.45
17	0.9997	62.34	4.00
18	1.0000	105.84	103.89

Note: Integrated age = 46.80 ± 0.8 (2σ) Ma.

TABLE 2. $^{40}Ar/^{39}Ar$ RESULTS OF HEATING UNDER VACUUM BY LASER OF THE SYLVITE SAMPLE

Step	Cumulative ^{39}Ar	Age (Ma)	±
1	0.0393	61.07	1.39
2	0.1212	19.59	0.74
3	0.1966	10.32	0.38
4	0.2784	16.70	0.50
5	0.3441	52.09	0.43
6	0.4198	37.64	0.56
7	0.4507	49.21	1.05
8	0.5599	53.52	0.63
9	0.6722	22.36	0.57
10	0.7876	28.74	0.51
11	0.8348	36.16	1.02
12	0.9178	46.88	1.02
13	1.0000	70.10	0.96

Note: Integrated age = 37.06 ± 0.80 (2σ) Ma.

TABLE 3. $^{87}Sr/^{86}Sr$ EXPERIMENTAL RESULTS OF SULFATES FROM THE EVAPORITE FORMATION

Sample	$^{87}Sr/^{86}Sr$	Age (Ma)
Basal anhydrite		
1	0.707831	33.84 ± 0.2
2	0.707802	34.57 ± 0.3
Anhydrite dispersed in halite		
3	0.707824	34.00 ± 0.2
4	0.707826	33.95 ± 0.2

Note: Numerical ages were obtained using the $^{87}Sr/^{86}Sr$ look-up table of Howarth and McArthur (1997).

tomeu del Grau sections, some of which are composed of several subsections (Fig. 2). These sections include the marine deposits of the basin, up to the overlying evaporites and continental deposits. The sampling sites were located using aerial photographs and the sections were measured with a conventional Jacob's staff.

Samples were collected using a portable gasoline-powered drill and were oriented in situ with a compass. The length of the cored samples was between 6 and 10 cm, in order to try to avoid any recent weathering effects. Two to four standard specimens of 2.54 cm diameter and 2.10 cm length from each sample were cut in the laboratory. Natural remanent magnetization (NRM) and remanence through demagnetization were measured on a double-shielded, three-axis, GM400 cryogenic magnetometer (CCL) at the Institute of Earth Sciences in Barcelona. This magnetometer has a noise level of 5×10^{-6} A/m, well below the intensity values of the studied samples. A Schonstedt TSD-1 oven with a residual field of 10 nT was used for thermal demagnetization. Alternating field (AF) demagnetization was performed on a Schonstedt GSD-5 AF tumbling-specimen demagnetizer. Orthogonal vector demagnetization plots (Zijderveld, 1967) were used to represent demagnetization data, and principal component analysis (Kirschvink, 1980) was used to determine the magnetization components.

Several lithologies (marine marls and sandstones) were sampled in a pilot study. After trials of both thermal (TH) and AF demagnetizations, it was found that the most suitable lithologies were marls and marly limestones, and therefore subsequent sampling was devised to incorporate sections where these units are dominant and correlate with the sandstone bodies in the northern and southern margins.

Paleomagnetic analysis was conducted on more than 500 samples, corresponding to 133 sampling sites from the southern margin (La Rovira, El Grau del Sunyer, Puigsagordi, Collsuspina, Tona) and 66 sites from the northern margin (Sant Bartomeu del Grau). Sampling encompasses a total vertical thickness of 1890 m. The average sampling interval was 6.5 m, although sites were more closely spaced in the uppermost marine units, where magnetic polarity intervals were expected to be shorter, according to the GPTS (Cande and Kent, 1992). An upper middle–upper Eocene age was previously proposed for these deposits by Serra-Kiel (1984) and Burbank et al. (1992a). At least two samples were taken per sampling site. However, the study was designed so that some sedimentary units were sampled several times across different sections, thus assuring the correct assignation to chrons, as well as correlation between proximal and distal facies. Resampling of several units from Burbank et al. (1992a) or their distal equivalents, have been used to compare to our results. Given the uncertainties in magnetic polarities shown by the coarser grained lithologies (Pascual et al., 1992; Parés and Dinarès-Turell, 1993), the sections El Grau del Sunyer and Puigsagordi and parts of the La Rovira and La Guardia sections were resampled after processing a first set of samples.

At least two specimens per site from different sampled cores were routinely subjected to stepwise thermal demagnetization, although AF demagnetization was used in a few cases. TH demagnetization involved between 8 and 12 steps, from room temperature to a temperature where the remanence intensity falls below the sensitivity of the magnetometer or until spurious behavior precluded further demagnetization. Intervals of 50 and 30 °C were used normally to a maximum temperature of 450–550 °C. AF demagnetization to 100 mT included a maximum of 13 steps with intervals of 5, 10, and 20 mT. The characteristic remanent magnetization (ChRM) is defined as the linear segment trending toward the origin of the demagnetization diagram. For the cases in which such a segment could not be well resolved, due to low intensities or spurious magnetizations acquired upon heating, the remagnetization circles technique (Halls, 1978; McFadden and McElhinny, 1988) was used to recover the remanence direction of the highest temperature (nonresolved) component. To obtain the site mean direction per sampled stratigraphic level, Fischer (1953) statistics were used to average the ChRM directions. For sites that presented both direct observations (end points determined from the linear segment trending toward the origin of the demagnetization diagram) and remagnetization circles for the ChRM component, the combination method of McFadden and McElhinny (1988) was followed to calculate the mean direction. Individual data are plotted in the declination and inclination logs together with the mean value and an additional quality criteria established as follows: class A for those sites that provide a minimum of two ChRM direction observations; class B if at least one ChRM observation was available; and class C if only remagnetization circles could be defined. The mean declination and inclination of the ChRM components at each site have been used to derive the latitude of the virtual geomagnetic pole (VGP). This parameter has been used as an indicator of the polarity (normal polarity for positive VGP latitudes and reverse polarity for negative VGP latitudes). The established magnetozones in the studied sections have been correlated to the geomagnetic polarity time scale (GPTS) of Cande and Kent (1995).

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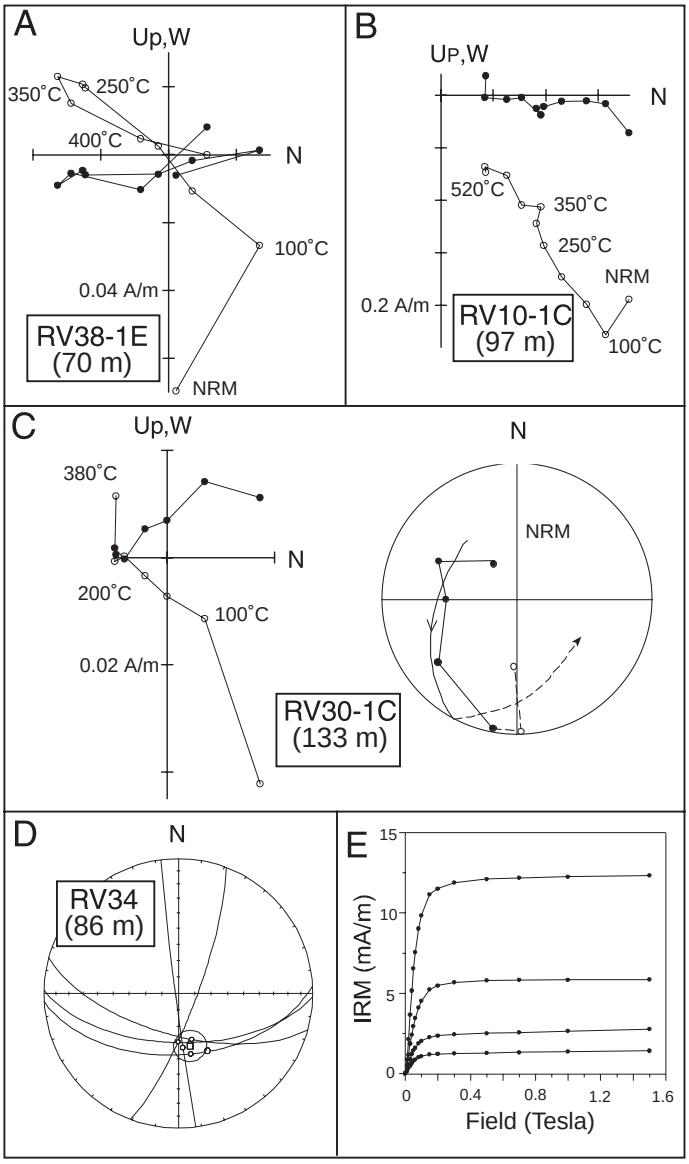


Figure 4. (A, B, C) Bedding-corrected representative orthogonal projections of thermal demagnetization data from the Vic area samples (section La Rovira, RV). Open and closed symbols represent projections on the vertical and horizontal planes, respectively. Temperatures in degrees Celsius for various points are indicated. Diagram C has been truncated at high temperatures, because intensity raises dramatically at those stages due to formation of new mineral phases upon heating. The high-temperature component in C is not well resolved, even though a reversed polarity is apparent and a remagnetization great circle was adjusted to sample RV30-1C. (D) Reliable determined directions as in A have been used as set points (large circles) to determine a best-fitting direction (small circles) from the great circles (after McFadden and McElhinny, 1988). (E) Examples of isothermal remanent magnetization (IRM) acquisition curves for representative samples. Samples reach saturation at about 0.2 T, indicating a low coercivity mineral as the magnetic carrier.

Although no fold test is available due to the nearly horizontal bedding in all sections, the following observations provide confidence that the calculated ChRM directions are accurate and primary in age: (1) the presence of geomagnetic reversals that correlate well between overlapping sections; (2) the presence of a low-temperature present-day field overprint in addition to the ChRM magnetization; (3) polarities do not appear to be lithologically controlled; (4) the mean inclination obtained is similar to the existing results in the area (Pascual et al., 1992; Parès and Dinarés-Turell, 1993).

PALEOMAGNETIC RESULTS: MAGNETOSTRATIGRAPHY

The intensity of the natural remanent magnetization (NRM) was moderate to weak, generally between 8×10^{-4} and 5×10^{-5} A/m in all sections. Two magnetization components can be recognized in most samples upon demagnetization, in addition to a viscous magnetization removed below about 100–150 °C (Figs. 4, 5, and 6). A low-temperature component (L) is unblocked between 150 and 300 °C and a high-temperature component (H) is unblocked usually above 300 °C until temperatures of about 450–500 °C. Component L has always a northerly declination and moderate to steep downward inclination (~60°) before rotation of bedding to the horizontal. The mean direction before tectonic correction for component L corresponds to the geocentric dipole direction for the present latitude of the sections and therefore can be related to a secondary recent overprint.

Typical examples of thermal demagnetization are illustrated in Figures 4, 5, and 6. A multicomponent character of the NRM is clearly shown in the majority of cases. In some instances (Figs. 4C and 5, A and C) samples show a spurious behavior at low temperatures, precluding the determination of a well-defined ChRM. In such cases component L does not trend to the origin of the demagnetization plot, implying the presence of a second magnetization component, and a remagnetization great circle has been fitted to recover the ChRM component.

The magnetic bulk susceptibility was monitored while heating the samples to control any mineralogical changes induced by oxidation in the oven during thermal demagnetization. Usually no significant change was detected below 420–450 °C. At this temperature susceptibility starts to increase substantially. This effect is commonly described in these types of sediments as a consequence of the formation of new, very fine grained magnetite (superparamagnetic) at the expense of the nonmagnetic iron sulfide (pyrite). This effect was more noticeable in those samples that exhibited spurious behavior at high temperatures. Attending the unblocking temperatures the carriers of the magnetization seem to be dominated by magnetite and/or maghemite, although a minor contribution from iron sulfides cannot be ruled out. Isothermal remanent magnetization (IRM) acquisition curves (Fig. 4E) show that saturation is reached below 0.2 T, suggesting a low coercivity mineral for the magnetic carriers. Comparable results from these types of sediments were obtained by Dinarès-Turell (1992). Subsequent thermal demagnetization of the acquired IRM usually indicates maximum unblocking temperatures of 500–550 °C, consistent with the presence of magnetite.

The computed ChRM components from the studied sections have been grouped and averaged for three arbitrary groups of sites (Figs. 2 and 7; Table 4). The southernmost sites include data from the La Rovira, El Grau del Sunyer, and Puigsagordi sections; the central sites comprise directions from the La Guardia, Tona, and Collsuspina sections; and the northernmost sites have the ChRM components from the different Sant Bartomeu del Grau subsections. We have performed the reversal test of McFadden and McElhinny (1990) in order to assess the antipolarity of the normal and reverse populations (Fig. 7). This test classifies a positive reversal test on the basis of the angle γ_c between the mean directions of the two sets of observations at which the null hypothesis of a common mean direction would be rejected with 95%

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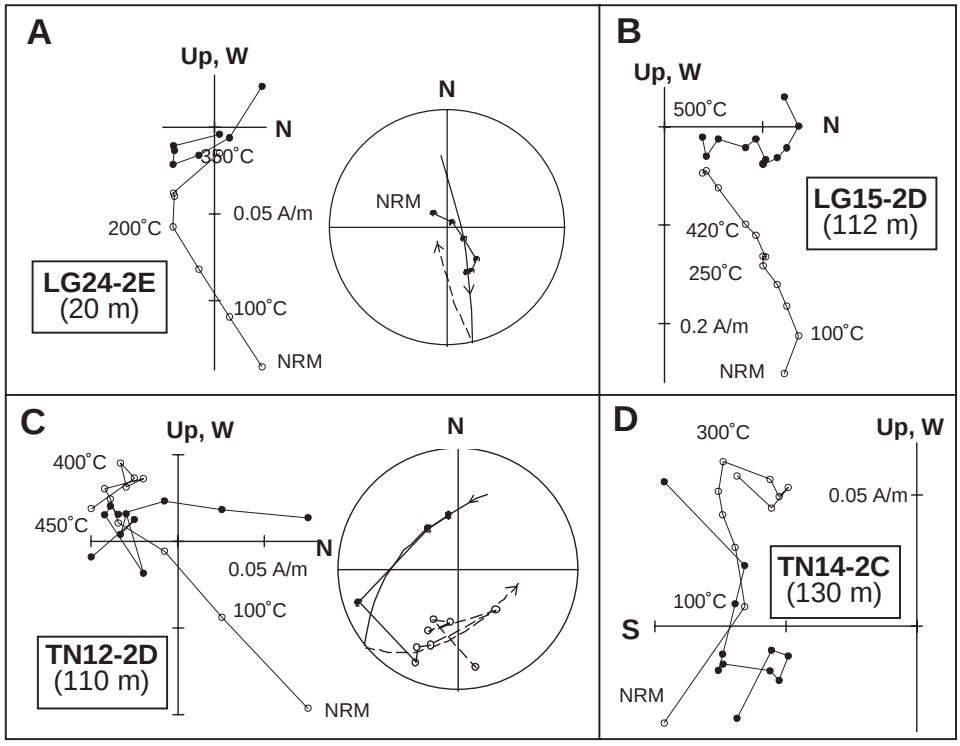


Figure 5. Bedding-corrected representative orthogonal projections of thermal demagnetization data from the Vic area samples (La Guardia [LG] and Tona [TN] sections). Open and closed symbols represent projections on the vertical and horizontal planes, respectively. Temperatures in degrees Celsius for various points are indicated. A remagnetization great circle was adjusted to samples LG24-2E (A) and TN12-2D (C).

confidence (class A if $\gamma_c \leq 5°$; B if $5° < \gamma_c \leq 10°$; C if $10° \leq \gamma_c \leq 20°$; and indeterminate if $\gamma_c > 20°$). Reversal type C has been obtained for the southern and central sites and type B for the northern sites.

Figure 8 (A–D) shows the VGP latitude curves for each study section. The ChRM mean direction for each site is an average for those with more than two samples. When only two data entries are represented, a moving average window by which the ChRM data for a particular site is averaged with the overlying and underlying site has been used. Under this scheme, adjacent sites with different polarity have been skipped for the individual respective mean. This allows the assessment of the coherency of the data and the assignment of an error estimate (e.g., the σ_{95} semiangle of confidence is calculated and plotted as an error envelope for each VGP latitude) (Fig. 8).

La Rovira and Grau del Sunyer Sections

These two sections are located at the southern margin of the southeastern Pyrenean foreland basin (Figs. 1B and 2A). The total thickness sampled at the La Rovira section is 445 m and samples most of the south-sourced deltaic units. The El Grau del Sunyer section (Fig. 2A) was sampled to gauge the validity of the lowest normal polarity zone (n1) found in the La Rovira section (Fig. 8A). Their magnetic polarity stratigraphy (MPS) contains eight magnetozones (Fig. 8A). The pattern of three long reversals (r1, r2, and r3) separated by two short normals (n1 and n2) suggests that the lowermost reversal corresponds to the chron C20r of the GPTS (Fig. 9). This is in good agreement with the age of $46.8 \pm 0.8(2\sigma)$ Ma obtained for the glauconite level at the base of the section. Given the pattern of reversals in this section, and the correlation with the other sampled sections throughout the basin (Fig. 9), we interpret the normal chron at the topmost part of the section to correspond to chron C18n.1n of the GPTS (Fig. 9). The time span represented by this section is about 6.5 m.y. (from base of chron C20r, ca. 46 Ma, to the base of chron C18n.1n, ca. 39.5 Ma). This gives an average ac-

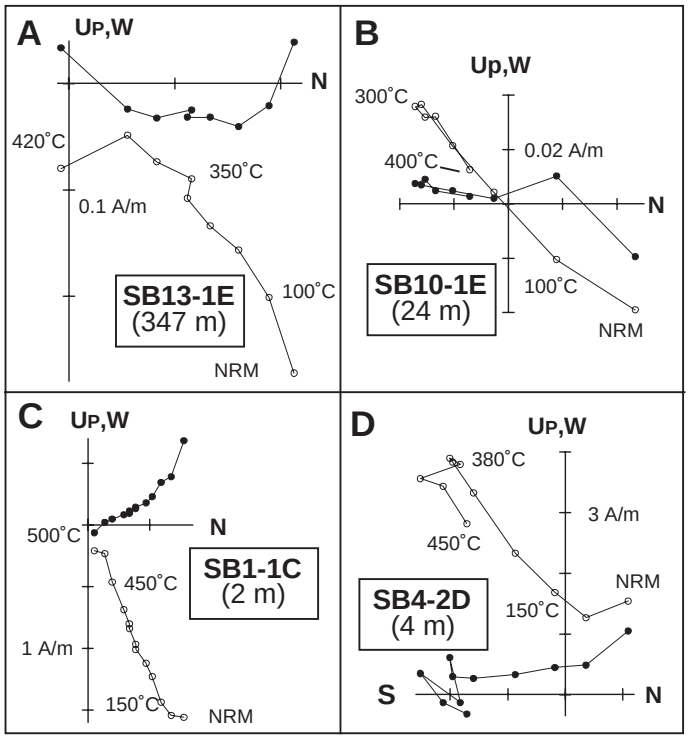


Figure 6. Bedding-corrected representative orthogonal projections of thermal demagnetization data from the Vic area samples (Sant Bartomeu del Grau [SB] section). Open and closed symbols represent projections on the vertical and horizontal planes, respectively. Temperatures in degrees Celsius for various points are indicated.

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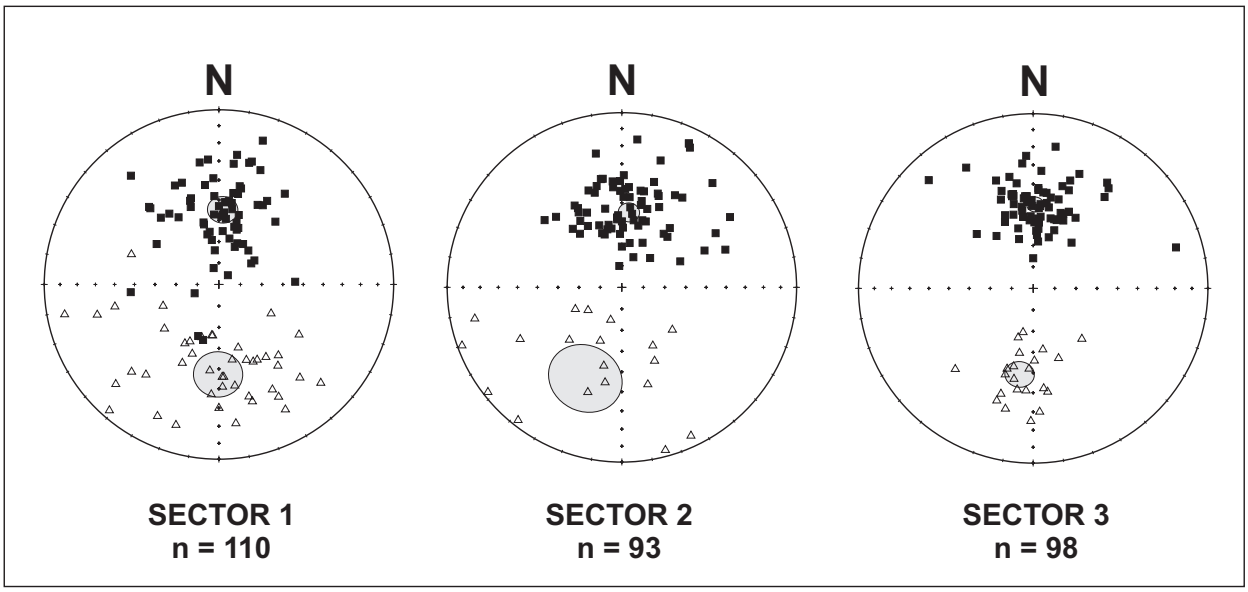


Figure 7. Equal-area projections of the characteristic remanent magnetization (ChRM) directions after tilt correction for the three arbitrary defined sectors in the Vic area (sector 1 includes La Rovira [RV], El Grau del Sunyer [GS], and Puigsagordi [PG] sections; sector 2 includes La Guardia [LG], Collsuspina [CS], and Tona [TN] sections; and sector 3 consists of Sant Bartomeu del Grau [SBG] section). Open and closed symbols represent projections on the upper and lower hemispheres, respectively. The mean directions for the normal and reverse populations and their confidence ovals are also shown (see Table 4).

cumulation rate for the entire sampled section of 0.07 m/k.y. before decompaction. The present correlation to the GPTS implies a significant increase in the sedimentation rate (0.33 m/k.y. before decompaction) during deposition of sediments representing r4. These sediments correspond to delta lobe deposits prograding on delta marls, which justifies the detected increase in the sedimentation rate.

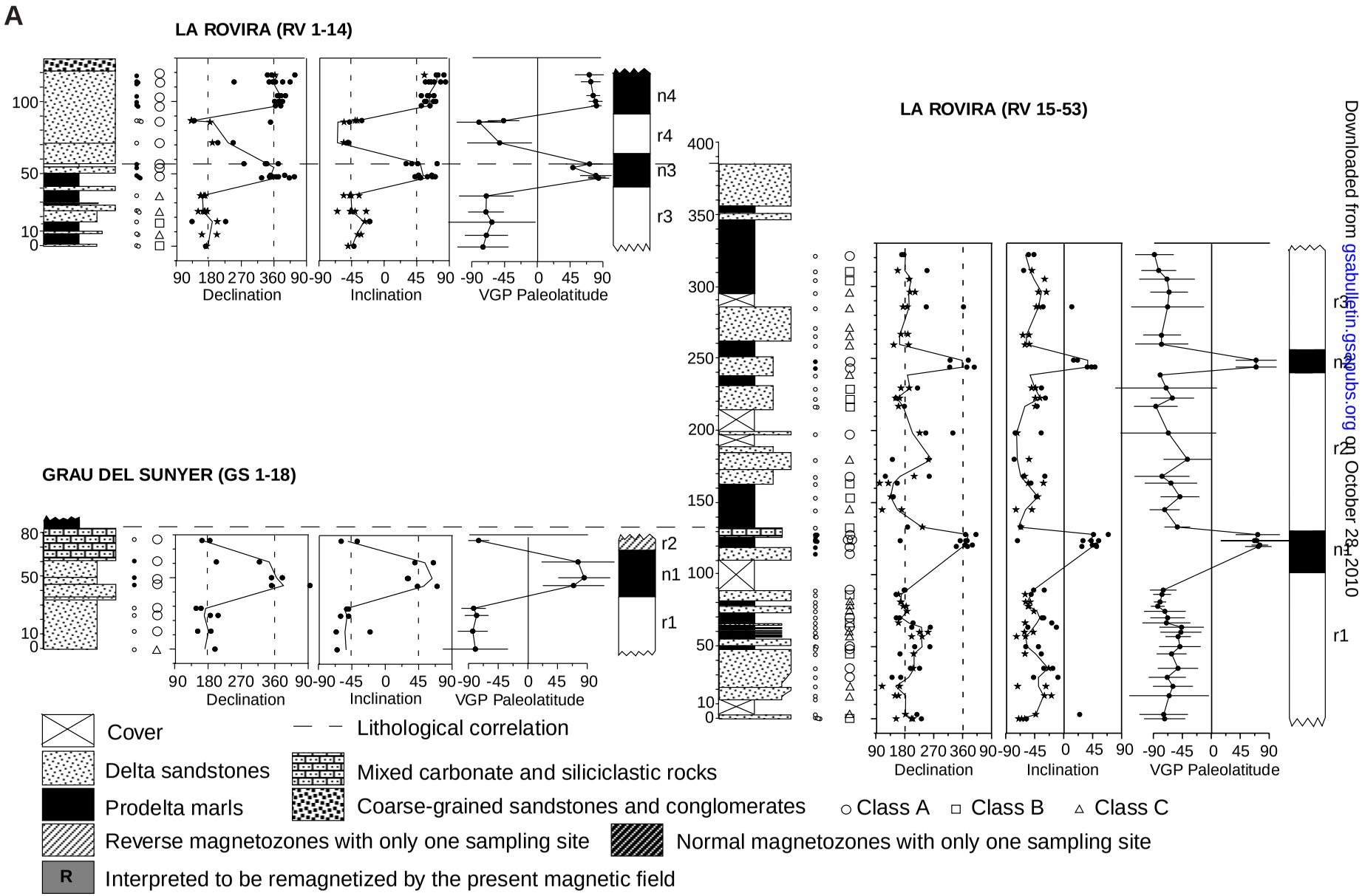
Puigsagordi Section

The objective of sampling this 130-m-thick section (Figs. 1B and 2A) was to aid in the correlation between the La Rovira and La Guardia sections. The MPS of this section (Fig. 8B) contains two magnetozones that are assigned to the GPTS by lithological correlation with the La Rovira section and the correlation of the upper reversal and the glauconite-rich deltaic top with the La Guardia section (2 and 2′ in Fig. 8B). We interpret the upper and lower magnetochrons present in this section to be C17r and C18n.1n, respectively. The lithological correlation with the La Guardia section enables the thickness of chron C17r in the Puigsagordi section to be interpreted. Lithological correlation of an erosion surface (1) in the lower part of the Puigsagordi section (Fig. 8B) with the surface marked as (1′) in the La Guardia section (Fig. 8B) proves that the lower part of the Puigsagordi section, which is composed of coarse-grained sediments, is remagnetized by the present magnetic field. However, the position of the boundary between chrons C18n.1n and C18n.1r can be deduced from the lithological correla-

TABLE 4. MEAN DIRECTIONS FOR THE CHARACTERISTIC REMANENT MAGNETIZATION (CHRM) OF THE VIC AREA

Polarity	N	Dec_b	Inc_b	α_{95}	k	Dec_a	Inc_a	α_{95}	k
Sector 1									
Normal	67	14.5	56.4	5.1	12.7	4.8	55.0	5.0	12.9
Reverse	43	189.9	–52.6	9.0	6.8	183.3	–49.5	9.0	6.9
All	110	12.7	55.0	4.6	9.6	4.2	52.6	4.6	9.6
Sector 2									
Normal	75	16.2	53.5	4.5	14.1	6.2	51.2	4.7	13.3
Reverse	18	215.0	–49.5	16.4	5.4	206.9	–47.6	16.9	5.2
All	93	19.7	53.0	4.7	10.6	10.0	50.9	4.9	10.0
Sector 3									
Normal	76	7.9	51.5	3.8	19.6	3.0	49.9	3.7	19.9
Reverse	22	190.7	–49.7	6.7	22.1	186.4	–47.8	6.4	24.1
All	98	8.6	51.1	3.3	20.2	3.8	49.4	3.2	20.8

**Notes:* Header abbreviations: N—Number of samples; Dec_b—Declination before bedding correction; Inc_b—Inclination before bedding correction; α_{95}—Half angle of the 95% confidence cone; K—Fisher's precision parameter; Dec_a—Declination after bedding correction; Inc_a—Inclination after bedding correction.



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Figure 8. The different magnetic polarity stratigraphy (MPS) and lithology logs of the studied sections: (A) La Rovira (RV) and Grau del Sunyer (GS); (B) Puigsagordi (PG) and La Guardia (LG), (1) is lithologically correlatable to (1′), and (2) is correlatable to (2′); (C) Collsuspina (CS) and Tona (TN), (3) is lithologically correlatable to (3′). From this point upward there is a remarkable increase in iron sulfide–rich layers. TCS—Tona Castle sandstones. (D) Sant Bartomeu del Grau (SBG). Individual data at each site are plotted as circles (direct reliable observations) and stars (fitted great circle direction) in the declination and inclination logs together with the mean curve as described in the text. The α_{95} error envelope is plotted for each virtual geomagnetic pole (VGP) paleolatitude.

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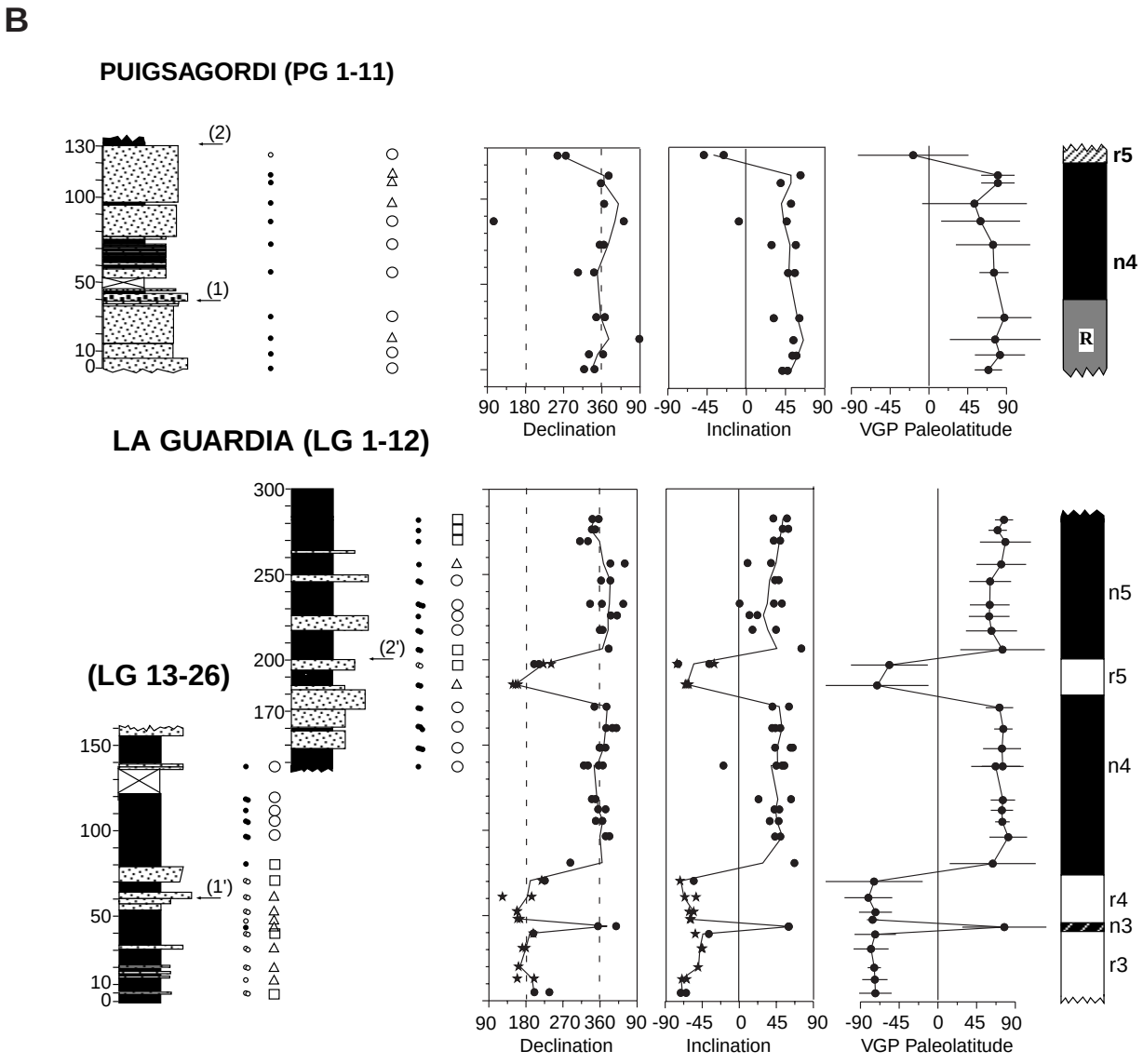


Figure 8. (*Continued*).

tion between both sections (Fig. 9). Thus, the time span represented by this section is calculated as ~1.2 m.y., yielding an average sedimentation rate for this section of ~0.11 m/k.y. before decompaction.

La Guardia Section

This section is located west of Els Hostalets (Fig. 2A); 290 m of sandstone units (delta lobes) and marl intervals (prodelta deposits) were sampled. The MPS of this section contains six magnetozones (Fig. 8B); n3 is represented by one site with two samples, and r5 is represented by two sites with a total of six samples. The magnetic features of each of these sampling sites, as well as the lithological correlations of marker beds with the La Rovira (RV1–RV14) and Puigsagordi sections and with magnetozone r5 of the Tona section (Fig. 9), verify that these are valid magnetozones, though only represented by a few samples. Following the scheme outlined here, the La Guardia section contains chrons C18r to C17n.1r, hence the time span of this section is ~3.5 m.y., implying an average sedimentation rate of ~0.08 m/k.y. before decompaction.

Collsuspina Section

The Collsuspina section is located close to the Vic-Manresa road in the Collsuspina area (Fig. 2A). The section is 50 m thick. Its lower part corresponds to prodelta marls, and its upper part was deposited under hypersaline anoxic conditions (Pierre et al., 1994). The MPS of this section shows two magnetozones (Fig. 8C), one of them represented by only one class C sampling site. The lithological correlation with the Tona section (Fig. 9) confirms the validity of the reversal zone at the bottom of the section. The pattern of a relatively long normal chron, spanning almost all the sampled section, and the correlation with the Tona section suggest that this chron is

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C

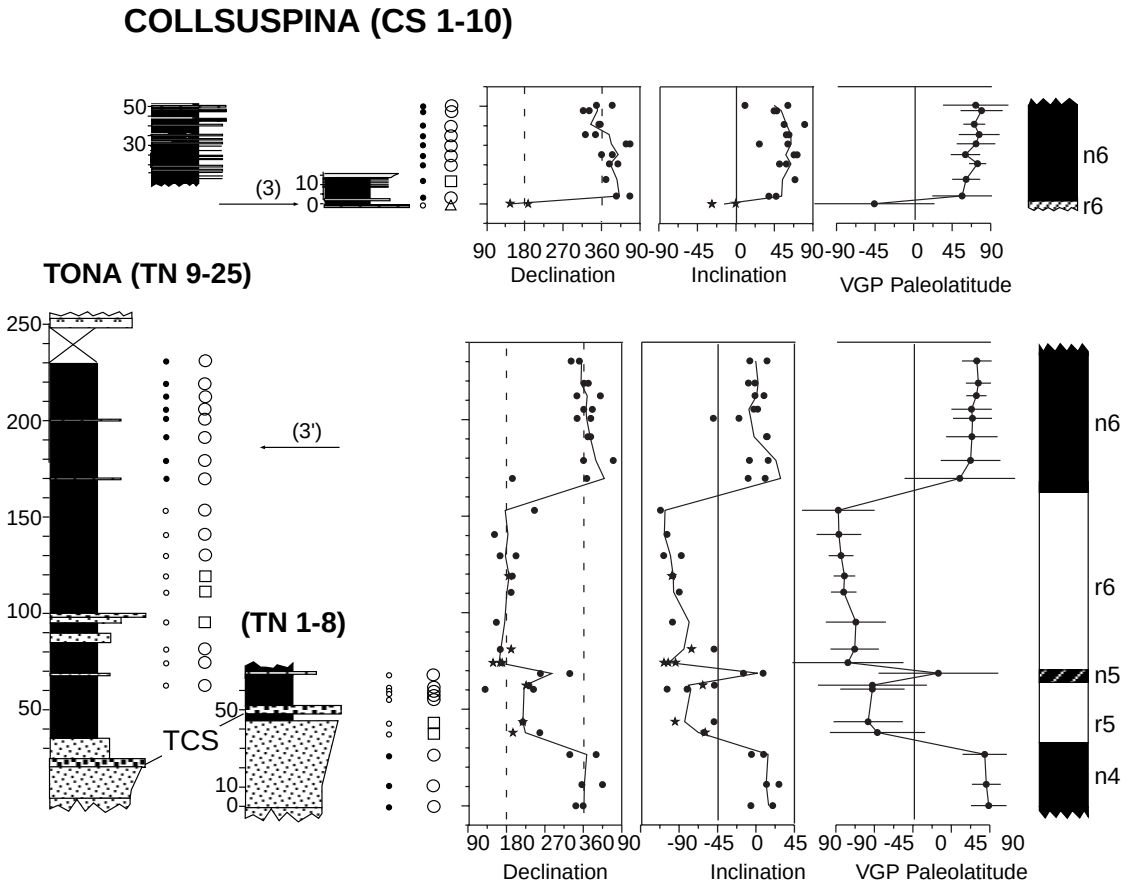


Figure 8. (*Continued*).

C17n.1n. The time span for this section is ~0.5 m.y., resulting in an average sedimentation rate of ~0.1 m/k.y. before decompaction.

Tona Section

The total sampled thickness of the Tona section is 230 m. The lowermost deposits sampled in this section correspond to delta lobe sandstones of the Centelles Formation; the sandstones are truncated by an erosion surface that can be traced southward to the glauconite-rich delta top in La Guardia and Puigsagordi sections (2 and 2′ in Fig. 8B). The first 40 m of marls above the sandstones that overlie the erosion surface are rich in sponges that are found in situ, colonizing scar-slump surfaces. This unit lithologically correlates with equivalent deposits of La Guardia and Sant Bartomeu del Grau sections (Fig. 9).

The MPS of this section (Fig. 8C) contains five magnetozones and their assignment to the GPTS is based on their reversal pattern and correlation with the southern sections. The normal magnetozone n6 (Figs. 8C and 9) is thus assigned to chron C17n.1n. This, together with the lithological correlation of the erosion surface in the lower part of Tona section to the delta top rich in glauconite in the La Guardia section (zone r5), assigns the other magnetozones present in the section to the GPTS. From this we can interpret that the lowermost normal zone of this section corresponds to chron C18n.1n. This implies that zone n5 incorporates chrons C17n.2n and C17n.3n; the short reversal between was not sampled and hence not detected. The minimum time span represented by this section is thus 1.5 m.y., which gives an average sedimentation rate of 0.15 m/k.y. before decompaction.

Sant Bartomeu del Grau Section

The Sant Bartomeu del Grau section is located in the northern sector of the basin (Fig. 8D). The total thickness sampled in this section is 665 m and the sediments (sandstones and marls) mostly correspond to north-sourced deltaic systems. The MPS of this section shows six distinct magnetozones (n4 to r8, Fig. 8D), which are correlated to the GPTS by the characteristic reversal pattern, the lithologic and magnetostratigraphic correlation with the Tona section, and the assignation of the Orís sandstones (Fig. 2B) to chron C18n.1n (in accordance with Burbank et al., 1992a). However, there could be some doubt as to the validity of the two magnetic zones shown at the top of the marine deposits in the Sant Bartomeu quarry section (samples 1–11), as there are discrepancies when comparing with the Sant Bartomeu road section (Fig. 8D). The samples from the quarry of Sant Bartomeu reveal that most of the anoxic marls above the reef talus deposits were deposited during a reverse period. It is likely that this was not detected in the Sant Bartomeu road section because it was very difficult to obtain good samples.

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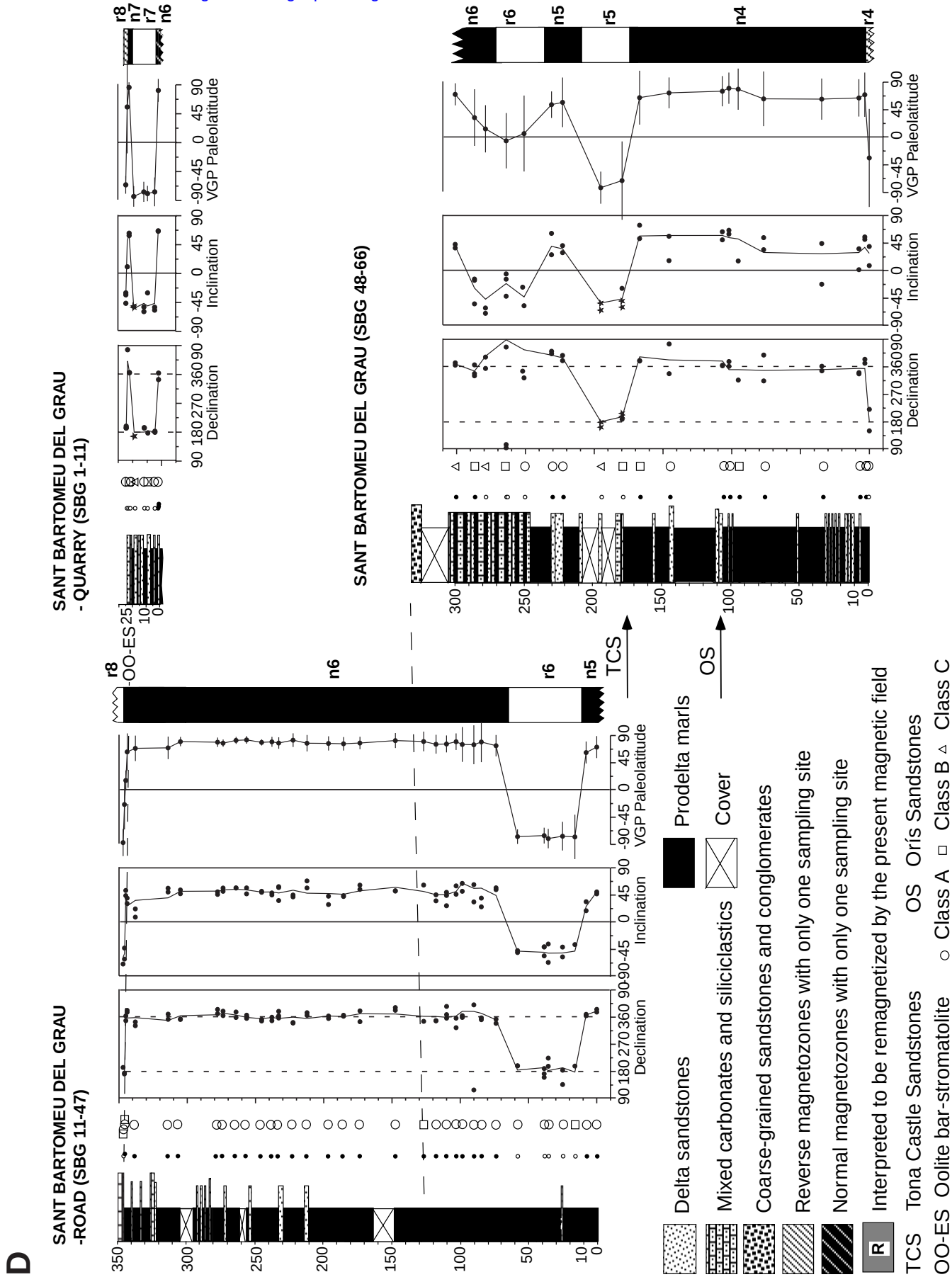


Figure 8. (*Continued*).

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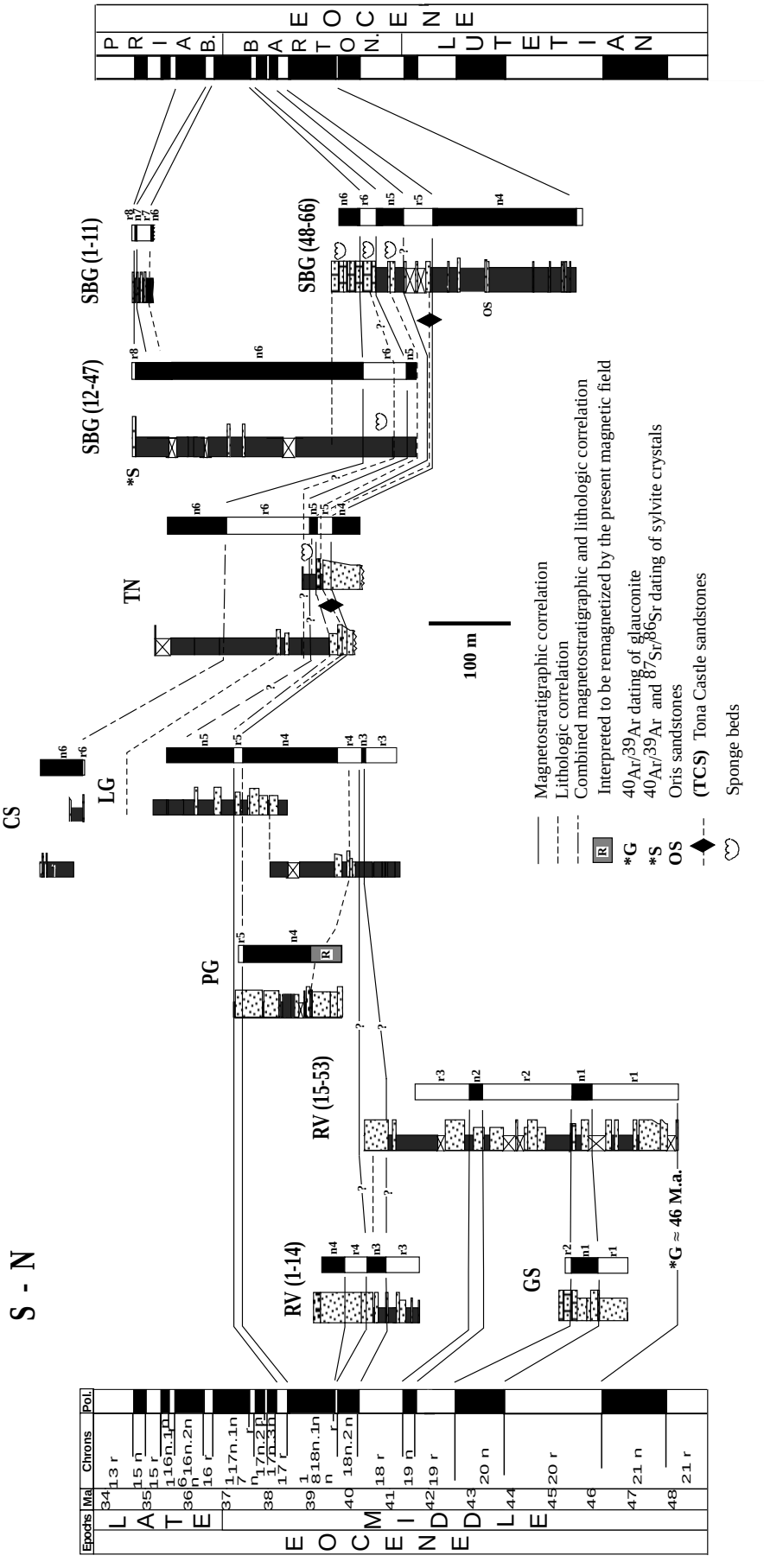


Figure 9. The magnetic polarity stratigraphy (MPS) across the Vic sector of the south-eastern Pyrenean foreland basin, correlated with the geomagnetic polarity time scale (GPTS) of Cande and Kent (1995). Lithostratigraphic correlation is also shown.

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TABLE 5. MAGNETOSTRATIGRAPHIC CORRELATION OF THE MPS OF THIS STUDY WITH THE GPTS

MPS (this study)	Chrons	Age (Ma)
r8	C16n.1r	35.68
n7	C16n.2n	36.34
r7	C16r	36.61
n6	C17n.1n	37.47
r6	C17n.1r	37.60
n5	C17n.2n	37.85
n5	C17n.2r*	37.92
n5	C17n.3n	38.11
r5	C17r	38.42
n4	C18n.1n	39.55
r4	C18n.1r	39.63
n3	C18n.2n	40.13
r3	C18r	41.25
n2	C19n	41.52
r2	C19r	42.53
n1	C20n	43.78
r1	C20r	46.26

Note: Ages are for the base of the chrons. GPTS from Cande and Kent (1995).
*C17n.2r is interpreted to be remagnetized or not detected in this study.

However, the polarity pattern at the very top of the section, displayed by the oolite bar and the stromatolites, is repeated in all of the outcrops where these beds have been sampled. Thus, the interpretation for the MPS pattern is that there are eight magnetozones recorded in this section.

Magnetozone n4 in the MPS of Sant Bartomeu del Grau is designated to chron C18n.1n of the GPTS, and the pattern for the entire section (Fig. 9) is consistent with the correlation of n6 to C17n.1n. We interpret that magnetozone r8 in the Sant Bartomeu del Grau section corresponds to chron C16n.1r, which would result in an age of 35.68 Ma for the stromatolite bed. This yields an age much younger than that obtained from the numerical dating of sylvite by $^{40}Ar/^{39}Ar$ and that suggested for the topmost marine deposits by others (Vergés and Burbank, 1996; Serra-Kiel et al., 1998). However, this age strongly correlates with the numerical age obtained from $^{87}Sr/^{86}Sr$ results from the evaporites. If we interpret the Sant Bartomeu del Grau section to begin in chron C18n.1n and finish in chron C16n.1r, this implies that this section represents a total time span of ~3.9 m.y., corresponding to an average sedimentation rate of ~0.2 m/k.y. before decompaction.

CORRELATION WITH THE GPTS ACROSS THE BASIN AND ITS IMPLICATIONS

The numerical age of the lowermost marine deposits in the southern sector of the basin (46 ± 0.8 Ma [2 σ] by $^{40}Ar/^{39}Ar$), together with the magnetic polarity reversal pattern of the La Rovira section (Figs. 8A and 9), implies that the onset of marine sedimentation correlates with the base of chron C20r at the southern sector of the basin, south of Centelles (Figs. 1B and 2A). This proves that the age of these marine deposits from the southern sector of the study area is considerably older (middle Lutetian), i.e., about 5 m.y. older than previously interpreted by Serra-Kiel and Travé (1995) and Vergés and Burbank (1996).

Because no major erosion surfaces were detected in the southern sector (La Rovira–El Grau del Sunyer sections) of the basin, it is assumed for the correlation with the GPTS (Table 5) that the observed reversal pattern can be assigned to the lowermost part of chron C20r at the bottom of the section, and to chron C18n.1n at the top (Fig. 9), given the radiometric ages outlined here. This pattern is in good agreement with the GPTS, with some minor differences: the sedimentary record of n1 is slightly thinner, and r4 thicker, than expected in the GPTS (Figs. 8, A and B, and 9). Nevertheless, this concurs with the sedimentological features of the sampled units. Reversal r4 has been assigned to C18n.1r (Table 5) and its relatively greater width implies a significant increase in the accumulation rate that is explained by a significant progradation of deltaic sediments consisting of coarse sandstones and conglomerates.

Some difficulties arise, however, when comparing the lower part of the Vic MPS from Burbank et al. (1992a) against our results from the southern margin of the basin and with the GPTS proposed by Cande and Kent (1992, 1995). The original work of Burbank et al. (1992a) used the Cenozoic geochronology proposed by Berggren et al. (1985), in turn based on Heirtzler et al. (1968), that proposed a model of two negative intervals within the positive anomaly 18. However, Cande and Kent (1992) modeled the interval around anomaly 18; they clearly indicated only one reversed interval that corresponds to the older of the two reversed subchrons recognized by Heirtzler et al. (1968) in chron C18n, but not the younger reversed subchron, which does not even amount to a tiny wiggle.

Vergés and Burbank (1996) assigned the MPS of Vic by linking to the GPTS proposed by Cande and Kent (1992), maintaining the previous correlation of Burbank et al. (1992a) to the GPTS. In doing so, Vergés and Burbank (1996) overcame the issue of correlating the second reverse subchron they found within C18n simply by regarding it as unmatched to the new Cande and Kent (1992) GPTS. We propose an alternative correlation to the GPTS for the reversal pattern below C18n.1n found in Burbank et al. (1992a) by simply shifting them down in the GPTS. This is based on the following: (1) In a similar position to this reversal we also have detected a reversal at the bottom of our Sant Bartomeu del Grau section (Fig. 8D), suggesting that this reversal is clearly represented in the northern margin of the basin. (2) The magnetic reversed polarity pattern of all sections sampled in our study, supported by numerical dating, suggests that in the Vic area chron C18n holds only one reversed subchron. We prefer to correlate the first reversal beneath C18n.1n as found by Burbank et al. (1992a), with C18n.1r (Table 5), thus solving the problem of correlation encountered in Vergés and Burbank (1996). Furthermore, we propose a direct correlation of the magnetozones found in Burbank et al. (1992a) with the Cande and Kent (1995) GPTS from chron C18n.1n downward to their zone N2. Their zone R2 then correlates with chron C19r, and their zone N2 most likely results from remagnetization under a (sub-recent) normal magnetic field. This zone (N2 of Burbank et al., 1992a) is only represented by a single site from a highly weathered permeable sandstone formation (Folgueroles Sandstone Formation, Fig. 3). If this were the case, N2 and R1 from Burbank et al. (1992a) would still represent chron C19r and the lowermost part of their Vic section would then still correspond to chron C20n, as these authors proposed, in order to tie with the available biostratigraphic data.

The correlation between sections sampled in the southern sector with those in the central and northern sector of the basin has been made possible by field mapping of a prominent marker unit (Tona Castle sandstones, Fig. 9) between the La Guardia and Sant Bartomeu del Grau sections. This sandstone unit developed on an erosion surface on top of delta lobe facies. The magnetic reversal polarity pattern encountered in the Tona sandstones, in the underlying deltaic sediments, and in marls with sponge beds is the same throughout the basin (Fig. 9). However, the variation in the width of the zones present in these sediments (r5, n5, r6) between different sections (Fig. 9) suggests that higher accumulation rates were controlled by depth and proximity to the sediment source areas in the south; i.e., proximal delta lobe facies are thicker than the equivalent distal marls.

When the sections that continuously sample the topmost part of the marine deposits are considered together with the lithological correlations be-

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tween them, a pattern of two wide normal polarity events (n4 and n6) with two short reversals separated by a short normal between can be observed (Figs. 8 and 9). The correlation of these zones with the GPTS suggests that n4 correlates with C18n.1n, in accordance with the interpretation of Burbank et al. (1992a) (their magnetozone N5). Then zone n6 may thus be considered to equate with C17n.1n in the GPTS. Combined lithological and magnetostratigraphic correlation proves that the width of n4 increases progressively from south to north (Fig. 9), demonstrating an increase in the accumulation rate in that direction.

Correlation of zone n6 to C17n.1n, together with the assignment of n4 to C18n.1n of the GPTS, implies that the span covered by zones r6, n5, and r5 in the basin must correlate with zones C17n.1r–C17r of the GPTS. This would indicate that one of the reversals of chron 17 is not registered in our sampled record. We propose that the reversal most likely not recorded in our sections is C17n.2r (Fig. 9, Table 5). This assumption is supported by the following observations: (1) this reversal is shorter than C17n.1r and C17r in the GPTS, and (2) this reversal is shown to be of little significance in some of the profiles used by Cande and Kent (1992) to constrain the widths of subintervals between anomalies 15 and 20.

Correlation of strata from north to south across the basin indicates that in late Bartonian time, the southeastern Pyrenean foreland basin was asymmetric and thickened to the north (Fig. 9), where more magnetozones are recorded. The Collsuspina and Tona sections terminate in a wide normal zone (assigned to C17n.1n) below the marginal evaporite formation, whereas the Sant Bartomeu del Grau section has two thin reversals separated by a very thin normal zone above this wide normal zone (Fig. 9).

The oolite bar–stromatolite pair (roughly equivalent to the lowermost outcropping evaporites) corresponds to the topmost sediments sampled (Fig. 8D). The oolite bar displays always normal polarity and the overlying stromatolite shows reversed polarity. The marls immediately below correspond to chron C16r in the Sant Bartomeu del Grau, but in the sections from the southern sector of the basin (Collsuspina, Tona, and in outcrops 6 km south of Sant Bartomeu del Grau) they correlate to chron C17n.1n (Fig. 9). Assuming that the oolite-stromatolite pair formed contemporaneously in the northern and southern margin of the evaporite basin, the existence of a sedimentary record that covers a larger time span beneath the evaporite equivalents at the northern sector (Sant Bartomeu del Grau area) implies nondeposition (or erosion and/or uplift) at the southern sector, while anoxic marine sedimentation continued at the north just before the entire basin became evaporitic. The topmost marine deposits below the evaporites are therefore not isochronous throughout the studied sector of the southeastern Pyrenean foreland basin. This implies asymmetry in the marine basin that is further demonstrated by the progressive increase in width of the magnetozone C17n.1n from south to north across the basin (Figs. 9 and 10).

Burbank et al. (1992a, 1992b) and Vergés and Burbank (1996), however, considered the transition between the marine sediments and the overlying continental series (1) to be time equivalent throughout the basin, (2) to correspond to a sea-level fall that coincides with the base of the evaporitic unit, and (3) to correlate with chron C17n.1n. According to the results of our magnetostratigraphic and lithostratigraphic correlation the evaporite unit is younger than the time span of chron C17n.1n. In addition, the boundary between the marine sequence and the overlying evaporites is diachronous, being located within chron C17n.1n in the northernmost sector of the study area (Burbank et al., 1992a), whereas in the relatively deeper areas at the center of the basin this boundary is located within chron C16r (Sant Bartomeu del Grau section). Furthermore, in the southern sector of the basin, the marls located just below the evaporites correspond to chron C17n.1n (Fig. 10). This diachroneity can be explained as a consequence of the basinward progradation of the deltaic systems combined with spatial variations between accumulation rate and subsidence in the basin. In relatively deeper areas of the basin (i.e., Sant Bartomeu del Grau section), a larger number of magnetozones is present below the evaporites (i.e., a larger time span is physically recorded) than at their marginal equivalents (Centelles section; Oliana section, Burbank et al., 1992b), where the evaporites are found on older sediments (chron C17n.1n).

DISCUSSION

Cross-basin magnetostratigraphic correlation aids in the interpretation of the evolution of sedimentary basins and can be particularly useful for shallow-water sediments where other means of correlation are not possible. The resolution that has been achieved for the southeastern Pyrenean foreland basin shows the advantages of this methodology to assess the evolution of architectural patterns in sedimentary basins through time and space.

The change from continental to marine sedimentation during middle Eocene time in this sector of the basin has been proven to be approximately isochronous, implying the record of a single significant event in the foreland with which this transgression is associated. Puigdefàbregas et al. (1986) interpreted this event to be the emplacement of the south Pyrenean thrust sheet units. Following the stratigraphy of the area available at the time, they proposed that the foreland depocenter migrated southward, causing strong diachroneity of the transgression. This complete scenario was deemed the consequence of load migration whereby depocenter migration was recorded between the Bellmunt, Milany, and Cardona sequences recognized by those authors (mainly corresponding to the deltaic deposits from the north to the evaporites; Figs. 3 and 10). These assumptions were based on the age constraints available at that time, i.e., assuming a Bartonian age (41.25 Ma when using the time scale of Cande and Kent, 1995) for the first marine deposits from the southern margin (Fig. 3). However, our results prove that the transgression at the bottom of the marine series is almost isochronous within the Vic sector of the basin. Marine sedimentation started around 46 Ma north and south of the Centelles high (Fig. 10).

Strong diachroneity is, however, still apparent when compared to other areas of the south Pyrenean foreland where the sediments from the southern margin crop out, such as the Igualada sector (Fig. 1A). Anadón et al. (1983) proposed that the lowermost marine Eocene deposits there correspond to zone NP17, on the basis of a study of the nannoplankton association. According to Berggren et al. (1995), NP17 comprises the top of C18r (40.13 Ma) to the middle part of C17n.1n (ca. 37 Ma). This implies that the transgression above the lower continental deposits is much younger at the southern margin of the basin in the Igualada sector. Vergés et al. (1998) interpreted that the southern margin of the foreland started its activity at 41.5 Ma and influenced the subsidence pattern in the basin. On the basis of these interpretations and the model of depocenter migration (Puigdefàbregas et al., 1986) at whole-basin scale, Vergés et al. (1998) proposed the migration of the "flexural wave" from the inner areas of the foreland to the most marginal areas in the southern edge of the foreland at the Igualada sector. However, the data from the Vic sector of the basin indicate that the activity in the southern margin of the foreland started at around 46 Ma. An alternative explanation for this difference in behavior between distinct sectors of the southern margin in the foreland is that it might be related to load migration along the Catalan Coastal Ranges (Fig. 1A), starting earlier in the northeastern side and propagating southwestward along the chain. We expect to gain new insights on this alternative explanation to "flexural wave" migration from the Pyrenees from subsidence analysis and flexural modeling studies currently in progress.

Although the transition between the lower continental deposits and the marine sequence occurred at about the same time in all areas of the studied sector, the lower continental and the older marine deposits are found to onlap, and eventually overlap, the Paleozoic and Mesozoic basement in the area around

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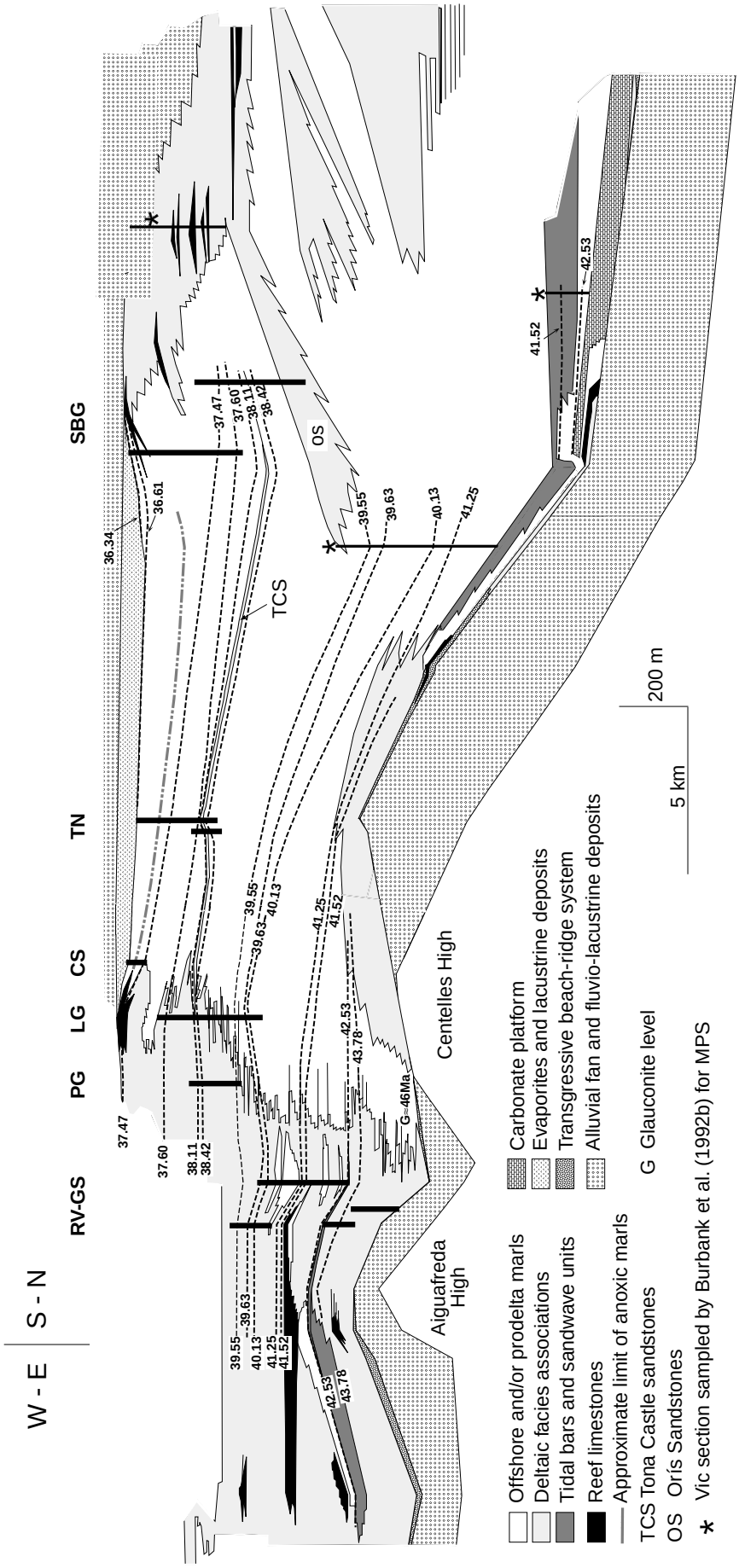


Figure 10. Cross-basin correlation of the easternmost southeastern Pyrenean foreland basin. Time lines show the variations through time and space of the infill architectural pattern and the correlation between north- and south-sourced depositional units in the basin. Numeric values of time lines (in Ma) represent boundaries of magnetic reversals after Cande and Kent (1995). The sections sampled and interpreted in this work are shown (La Rovira–El Grau del Sunyer [RV-GS], Puigsagordi [PG], La Guardia [LG], Collsuspina [CS], Tona [TN], and Sant Bartomeu del Grau [SBG]), as well as the location of the Vic section (*) sampled by Burbank et al. (1992a). Age values for the lower part of Burbank et al. (1992a) section have been repositioned according to the discussion in the text. MPS—magnetic polarity stratigraphy.

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the Centelles high (Figs. 1B and 10). This may imply that the high is draped-over basement topography. However, the stratigraphy presented here (Fig. 10) indicates that the area where the Centelles high is located might have corresponded to a structure that actively controlled sedimentation. (1) There are two principal subbasins north and south of the Centelles high for most of the marine sequence (to 38.11 Ma). (2) The variation in thicknesses of sediments accumulated between time lines north and south of the Centelles high, at least to 38.42 Ma, suggests the changing of depocenters between the areas north and south of the Centelles high (Fig. 10).

The infill patterns displayed by the evolution of sedimentary facies, together with the time lines provided by this study, demonstrate accommodation of sediment by aggradation up to an equilibrium represented by the base level in the southernmost sector of the study area between the La Rovira–El Grau del Sunyer sections and the La Guardia and Collsuspina sections. According to De Celles and Giles (1996), this architectural style is typical of backbulge depozones in foreland basins. The thinning of the sedimentary packages, as well as the lateral shift with time between depozones in the area of the Centelles high, shows similarities to those displayed in the schemes of evolution of the forebulge area in the Appalachian foreland basin (Tankard, 1986), although on a much smaller scale. This geometry suggests that the Centelles high might correspond to a forebulge that separates a backbulge area, sediment supply being from the main foreland, from a main depocenter supplied from the Pyrenees. The area that might be interpreted as a forebulge is very narrow (5 km) in the study sector of the basin (Figs. 1 and 10), although it widens to the east in the Montseny massif, which is 1700 m high. In a typical flexural basin the forebulge should be on the order of 60–470 km wide and a few tens to a few hundred meters high (De Celles and Giles, 1996). However, broken plates and plates with lower flexural rigidity should have higher and narrower forebulges than infinite plates and more rigid plates (Crampton and Allen, 1995; De Celles and Giles, 1996). Work in progress on flexural modeling based on the geohistory across the studied sector of the foreland will provide constraints for (1) the possible interpretation of the Centelles high as a true forebulge, (2) the understanding of the causes for changing of depocenters, and (3) the refinement of the history of thrust emplacement in the Pyrenees as well as in the Catalan Coastal Ranges as recorded in the infill architecture of the foreland.

The new results for the sediments studied at the southern sector of the Vic area of the southeastern Pyrenean foreland basin (south of Centelles high) provide a much simpler model for the architecture of the basin infill, recording almost identical trends throughout the study area. (1) The transgression at the bottom of the marine sequence can be considered to be essentially a single and time-equivalent episode. (2) Time-equivalent units were formed in similar depositional environments. A good example of the latter is shown by the development of tidal bars and sandwave units at almost the same time on both sides of the Centelles uplift (ca. 42.53 Ma, Fig. 10). The same trend is also shown by the sedimentation of deltaic units associated with coral reefs. However, despite the general trend, the correlation across the basin (Fig. 10) has proved that similar type facies developing in both north and south sectors of the basin are not exactly isochronous (i.e., reefs on abandoned delta lobes, the anoxic marl unit at the top of the studied marine sequence, the sediments below the evaporites, or the marine to continental transition) in different sectors of the basin (Fig. 10).

The correlation across the studied sector of the southeastern Pyrenean foreland basin shows that the basin infill was approximately symmetrical from the time line representing the progradation of the deltaic bodies of the Orís sandstones in the marine section to the restriction in the basin and evaporite precipitation (Figs. 3 and 10). However, the time correlation across the basin (Fig. 10) reveals (1) northward displacement of the depocenter at the top of the marine section in the north of the study area, close to our Sant Bartomeu del Grau section, as well as (2) the hiatus and/or possible emergence of the preevaporite marls at the southern margin of the basin while anoxic marl sedimentation still continued in the northern margin prior to evaporite precipitation. This suggests that the marine platforms from the southern margin were emergent while marine sedimentation continued in the north prior to evaporite sedimentation.

The accumulation rates changed through time and space across the basin, with a general increase in the average rate of accumulation from south to north and from 41.52 Ma upward in the marine series in the northern half of the basin. However, the accumulation rate is much higher south of Centelles up to about the 41.52 Ma time line. This explains the distribution pattern of the deltaic facies: delta lobes are stacked vertically south of the Centelles high, while the thickness of shallow platform facies is reduced north of Centelles (Fig. 10). However, the accumulation rate is observed to change across the basin after about 41.52 Ma: the thickness and type of sediments recorded north of Centelles imply a significant subsidence of the basin, probably related to the emplacement of the south Pyrenean thrust sheets. However, this event had only a slight effect south of Centelles, where the depth of deposition decreased with time after an initial deepening, and delta lobe deposits prograded from the southern margin of the basin up to the latitude of Centelles (Figs. 1B, 2A, and 10). There, the deltaic lobes once again stack vertically to the top of the marine sequence (Fig. 10).

For the sections analyzed in this study the average undecompacted accumulation rate varies from 0.07 m/k.y. (La Rovira section) to 0.2 m/k.y. (Sant Bartomeu del Grau section), and within any particular section the rate of accumulation varies through time. It is clear that any subsidence analyses, tectonic interpretations, or the timing of deformation from a single section would give rise to only a part of the geologic history. The results obtained for the studied marine deposits from the selected area of the southeastern Pyrenean foreland basin provide estimates of the rates of sedimentation and subsidence across the basin and implications for their controls. The evolution of the architectural infill in the foreland from the study of the distribution of the time lines across the studied area is observed to be complex and variable over the 40 km represented in the cross section of Figure 10. This reveals that, to understand the causes of evolution of this foreland and its relation to the nearby orogens (Pyrenees and Catalan Coastal Ranges), it is essential to analyze these cross-basinal variations. This is possibly the case for many other foreland basins where the magnetostratigraphic study of a single section may account only for part of the sedimentary basin infill history, and resulting subsidence analyses might only be valid for that particular section.

CONCLUSIONS

Magnetostratigraphy, constrained by numerical dating and supported by facies mapping, provides a reliable tool for cross-basin correlation. This approach overcomes the problems of misinterpretation due to missing key magnetozones by, or due to, remagnetized beds, as laterally equivalent formations are sampled several times in sections throughout the basin. On the first occasion that this methodology has been applied to the south Pyrenean foreland basin, the following conclusions may be postulated. (1) The marine sediments from the Eocene southeastern Pyrenean foreland basin are about 5 m.y. older than has been previously interpreted, the base of the marine sequence being middle Lutetian (ca. 46 Ma). (2) An age of about 35 Ma is proposed for the Cardona evaporite unit and thus the total time taken to form the marine deposits of the southeastern Pyrenean foreland basin in the Vic sector was ~10 m.y. (3) The second polarity reversal in chron C18 (C18n.2r) of Berggren et al. (1985) is not observed in the Vic sector. (4) There is no evidence to support a significant southward migration of the southeastern Pyrenean foreland basin depocenter at the Vic sector during the initial stages of the marine sequence sedimentation. (5) Subsidence and rate of accumulation

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changed in time and space across the basin. (6) The stratigraphy supports depocenter switching on either side of the Centelles high. (7) A preliminary interpretation for the Centelles high corresponding to a forebulge structure suggests that the south Pyrenean foreland basin might be a good example where sedimentation in the backbulge area is well documented. (8) Asymmetry within the latest basin infill sequence is indicated by a northward increase in the thickness of magnetozone C17n.1n and diachroneity within the marine deposits beneath the evaporites (although the onset of evaporite precipitation was isochronous for the studied sector). (9) Any correlation within the basin that considers the marine deposits below the evaporites to be isochronous is likely to result in misinterpretation of basin geohistory.

These findings redefine the basin infill architecture and the age calibration used by geologists to estimate rates in tectonic processes and sedimentation within the southeastern Pyrenean foreland basin. In particular, this basin has been used as a model for demonstrating the migration of depocenters in foreland basins in front of the orogen as a consequence of the migration of frontal thrusts (Puigdefàbregas et al., 1986). This study proves that this was not the case for the studied sector of the basin, where Puigdefàbregas et al. (1986) were supporting their hypothesis. Depocenter migration might, however, have existed on a larger scale in this basin, as suggested by Vergés et al. (1998). However, an alternative explanation for the relatively significant accumulation rates recorded at the earlier stages of deposition of the marine sequence at the southern margin of the basin (either in the Vic or Igualada sectors) is that this is a response to the activity in the Catalan Coastal Ranges.

The approach followed provides a reliable and robust means of basin-wide correlation. This suggests that for the studied basin, and possibly other foreland basins, a study with objectives similar to that presented here is necessary before developing further basin analysis, such as subsidence evolution in relation to tectonic events or thrust emplacement, or quantifying the parameters controlling the infill architecture by computer modeling. All information on sediment infill and the time correlations between sedimentary units are required before further conclusions on the evolution of a foreland basin might be proposed.

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